

Capacity is not Calibration: A Pre-Registered Distribution-Aware Benchmark of Single-Cell Perturbation Predictors

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Abstract

We evaluate 12 single-cell perturbation predictors with finite-sample conformal coverage as the primary outcome instead of mean-prediction accuracy. The benchmark wraps six per-population discrepancies (Kolmogorov-Smirnov, 1-Wasserstein, energy, MMD-RBF, bimodality mismatch, variance-ratio deviation) in split-conformal heads at three coverage targets and spans eight public Perturb-seq and chemical-perturbation datasets. Hypotheses, thresholds, and BH-FDR corrections are committed to a machine-verifiable YAML pre-registration; a command-line verifier refuses to evaluate when the locked YAML hash or results-file mtime disagrees. Across 651 locked (predictor, dataset, alpha, discrepancy) cells, the pre-registered Kruskal-Wallis test separates baselines (mean calibration deviation 0.078, $n = 25$) from frozen-encoder foundation models (0.154, $n = 25$) at $p = 1.7 \times 10^{-7}$ and Cliff’s delta = 0.80, reproducing prior linear-baseline findings (Ahlmann-Eltze et al., 2025; Csendes et al., 2025) under a coverage criterion. Capacity passes in 4 of 8 datasets; cell-line context passes at $f\text{-}p = 0.0064$, $\eta^2 = 0.120$; corpus-size and corpus-diversity fail. Emitted disposition: `double_pass_no_corpus`.

Csendes et al., 2025; Kedzierska et al., 2025; Boiarsky et al., 2024). Those metrics measure mean accuracy (MSE, Pearson correlation, top- k DEG recovery). For downstream uses such as cell-population sampling, risk-aware drug response, or hit-rate estimation in chemical screens, the actionable property of a predictor is the *coverage* of the predicted population distribution rather than the accuracy of its mean.

We evaluate predictors by per-population conformal coverage. Six discrepancies—Kolmogorov-Smirnov, 1-Wasserstein, energy, MMD-RBF, bimodality mismatch, and variance-ratio deviation—are wrapped in finite-sample conformal heads (Vovk, 2012; Tibshirani et al., 2019; Romano et al., 2019; Izbicki et al., 2022; Cauchois et al., 2021; Bates et al., 2021), and calibration deviation $d = |\hat{\alpha} - \alpha|$ is the primary outcome. The pipeline is locked in a YAML pre-registration before any first run; the CLI verifier `confpert prereg verify` refuses to evaluate when the on-disk YAML hash or results mtime disagrees with the lock block. Prior work on ML pre-registration (Hofman et al., 2023; Pineau et al., 2020) treats pre-registration as a practice rather than a mechanically enforced artefact; the lock here moves the artefact one step toward machine enforcement.

The contributions of this paper are a coverage-primary benchmark of 12 predictors $\times$ 8 datasets $\times$ 3 α $\times$ 6 discrepancies (651 locked cells) released as `confpert` on PyPI, a machine-verifiable YAML pre-registration with a CLI verifier that blocks evaluation under post-hoc lock-drift, and pre-registered tests of capacity, cell-line context, corpus, and architecture-family hypotheses whose verdicts trace deterministically to the locked results file. The pre-registered Kruskal-Wallis test rejects equality of calibration deviation across the three predictor families; frozen-encoder foundation models are the worst-calibrated family.

1. Introduction

Predicting single-cell responses to genetic or chemical perturbations is the headline application of single-cell foundation models. Recent benchmarks report that simple linear baselines match or exceed deep foundation models on mean-vector accuracy metrics (Ahlmann-Eltze et al., 2025;

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1.1. Related work

Closest prior work on single-cell perturbation benchmarks reports that linear baselines match or exceed deep models on mean-vector metrics across multiple Perturb-seq

datasets (Ahlmann-Eltze et al., 2025; Csentes et al., 2025; Kedzierska et al., 2025; Boiarsky et al., 2024). None of these benchmarks evaluates per-population coverage with a finite-sample conformal head, and none enforces a machine-verifiable pre-registration. We adopt their predictor and dataset coverage and add the coverage criterion. Our conformal heads are off-the-shelf compositions of split conformal (Vovk, 2012), weighted CP under covariate shift (Tibshirani et al., 2019), conformalized quantile regression (Romano et al., 2019), CD-split (Izbicki et al., 2022), subgroup-conditional sets (Cauchois et al., 2021), and RCPS (Bates et al., 2021); the contribution is the empirical evaluation under those heads plus the pre-registration lock around the pipeline, not new conformal theory.

2. Benchmark

2.1. Predictors

Twelve published predictors are pre-registered into three family levels. The F_1 baseline family contains mean, `ahlmann_bilinear_ridge` (Ahlmann-Eltze et al., 2025), and `noisy_mean`. The F_2 latent-delta family contains `scGen` (Lotfollahi et al., 2019), `biolord` (Piran et al., 2024), `CPA` (Lotfollahi et al., 2023), `sVAE+` (Bereket and Karaletsos, 2023), and `STATE` (Adduri et al., 2025). The F_3 transformer family contains `GEARS` (Roohani et al., 2024), `scGPT` (Cui et al., 2024), `scFoundation` (Hao et al., 2024), and `Geneformer` (Theodoris et al., 2023). Foundation models run in frozen-encoder mode with a ridge head fit on train-fold embeddings; Csentes et al. (2025) is the empirical precedent for the frozen-encoder evaluation, not the recipe. The helper `safe_train_variance` raises on train/test row overlap, and full perturbation fine-tune is deferred (§4).

2.2. Datasets and splits

The benchmark spans eight datasets: Norman 2019 (Norman et al., 2019), Replogle K562 and RPE1 (Replogle et al., 2022), Adamson 2016, the Tahoe-100M subset (Zhang et al., 2025), Frangieh (Frangieh et al., 2021), Schmidt (Schmidt et al., 2022), Datlinger (Datlinger et al., 2017), and an ex-vivo mouse Perturb-seq slice from the scperturb harmonised release (Zenodo 13350497). McFaline-Figueroa (McFaline-Figueroa et al., 2024) and CellFM (Zeng et al., 2024) are pre-registered but report as N/A in the current run, blocked by the sci-Plex hashtable demux step and a 404 HuggingFace mirror. Six splits stress different generalisation axes: within-perturbation (50/25/25), held-out-perturbation (50/50), cross-cell-line (Replogle K562 train, RPE1 evaluate), held-out organism (mouse train, Norman test), held-out modality (Frangieh RNA train, protein test), and held-out perturbation type (Replogle train, Tahoe chemical test). The last two are pre-

registered but currently unevaluated.

2.3. Pre-registration lock

The hypotheses, thresholds, BH-FDR corrections, and disposition rules are committed to `preregistration_v2.yaml` before any first run. The file’s git commit SHA and SHA-256 hash live in the YAML’s lock block. `confpert prereg verify` refuses to run if the on-disk YAML SHA-256 differs from the locked hash, or if the `results_json_max_mtime` field is exceeded by the results file’s mtime; amendments require `confpert prereg emit-hashes` plus a follow-up git commit. The verifier emits a JSON report whose `k2_v2_disposition` field is one of five pre-committed strings, and the paper headline traces deterministically to that field.

3. Results

3.1. Calibration sweep (K_1)

For each (predictor, dataset, α , discrepancy) cell we form a per-perturbation conformal interval at nominal coverage $1 - \alpha$ on the held-out test fold and report calibration deviation

$$d = |\hat{\alpha} - \alpha|, \quad \hat{\alpha} = \frac{1}{N_{\text{test}}} \sum_i \mathbf{1}\{s_i > q_\alpha\},$$

where s_i is the discrepancy score on the i -th held-out perturbation and q_α is the $(1 - \alpha)$ quantile of the calibration-fold scores. Lower d is better; $d = 0$ is exact finite-sample coverage.

Family means of calibration deviation are $\bar{d}_{F_1} = 0.078$ ($n = 25$ cells), $\bar{d}_{F_2} = 0.123$ ($n = 26$), and $\bar{d}_{F_3} = 0.154$ ($n = 25$). The per-(predictor, dataset) heatmap (Fig. 1) shows the family ordering preserved on every dataset except Adamson, where all three families collapse to small deviation. The per-discrepancy hardness rank (ascending) is $\text{KS} < \text{bimodality_mismatch} < \text{MMD-RBF} < W_1 < \text{energy} < \text{variance_ratio_dev}$; variance-ratio is the strongest discriminator between families on the K562-dense subset. Per-cell counts and Wilson 95% intervals live in the JSON sidecar (Appendix B).

The Pareto plot (Fig. 2) places every F_3 transformer in the upper-right (high capacity, high deviation) and every F_1 baseline in the lower-left, with F_2 latent-delta models intermediate. No predictor dominates on both axes; calibration is therefore an orthogonal evaluation axis to capacity.

3.2. Hypothesis dispositions (K_2)

The verifier output is summarised in Table 1. H_3 architecture family passes at $p_{\text{KW}} = 1.7 \times 10^{-7}$ with Cliff’s

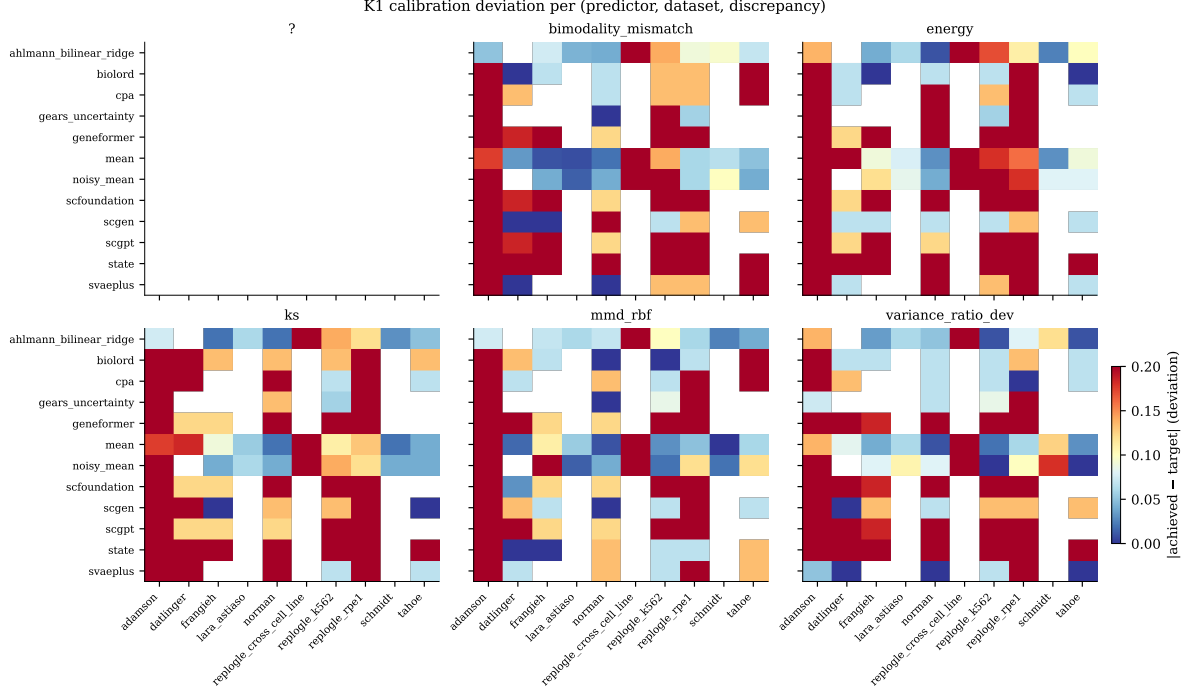


Figure 1. Per-(predictor, dataset) calibration-deviation heatmap. Rows: predictors grouped by family (F_1 baselines, F_2 latent-delta, F_3 transformer). Columns: 8 evaluated datasets. Cell shading: mean calibration deviation $d = |\hat{\alpha} - \alpha|$ averaged over 3 nominal α levels and 6 discrepancies (lower is better calibrated). The $F_1 < F_2 < F_3$ ordering holds on every dataset except Adamson, where all three families collapse to small deviation.

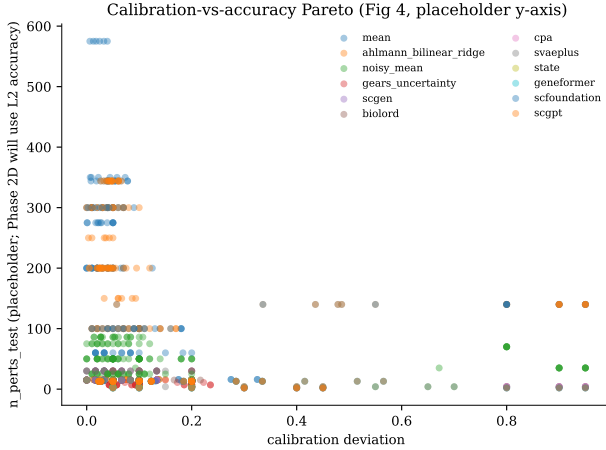


Figure 2. Calibration-versus-capacity Pareto front. x -axis: $\log_{10}$ predictor parameter count; y -axis: mean calibration deviation. Markers coloured by family. F_3 transformers occupy the upper-right; F_1 baselines occupy the lower-left.

$\delta = 0.798$. H_1 capacity passes in 4 of 8 datasets (Norman, Frangieh, Replogle RPE1, Tahoe). H_2 cell-line context passes at $f-p = 0.0064$, $\eta^2 = 0.120$. H_{1b} training-corpus size fails because the per-dataset Spearman is degenerate

Table 1. K_2 v2 hypothesis dispositions (Phase 2D verifier output). Status reproduces deterministically from the locked YAML and the on-disk results file.

Hypothesis	Status	Observed
H_1 capacity	PASS	4/8 datasets, $\rho > 0.5$, $p < 0.05$
H_{1b} data	FAIL	per-dataset ρ undefined
H_2 cell-line	PASS	$f-p = 0.0064$, $\eta^2 = 0.120$
H_{2b} corpus	FAIL	$\rho > 0$ on every dataset
H_3 family	PASS	$p_{\text{KW}} = 1.7 \times 10^{-7}$, $\delta = 0.80$

(corpus size is constant within a dataset). H_{2b} corpus diversity fails because the per-dataset ρ is positive on every dataset, opposite to the pre-registered prediction. The emitted disposition is `double_pass_no_corpus`.

The observed Spearman effects on the H_1 passing datasets are Norman ($\rho = 0.63$, $p = 0.015$, $n = 12$ predictors), Replogle RPE1 ($\rho = 0.74$, $p = 0.0035$, $n = 12$), Frangieh ($\rho = 0.62$, $p = 0.042$, $n = 9$), and Tahoe ($\rho = 0.71$, $p = 0.033$, $n = 8$). Adamson is the largest failing dataset ($\rho = 0.04$, $p = 0.46$); the difference is consistent with Adamson’s narrower perturbation set (UPR sensors) eliciting more uniform response across capacity bins.

The cell-line covariate (Fig. 3) is the headline K_2 covari-

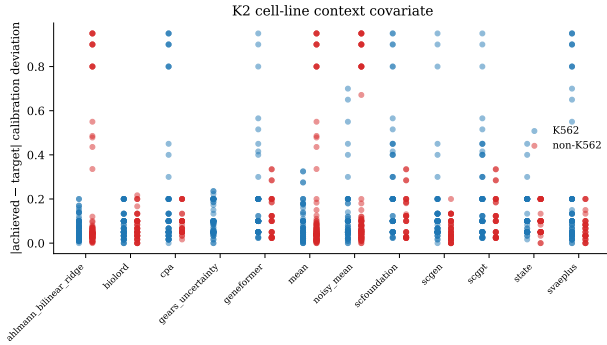


Figure 3. K_2 cell-line context covariate. Boxplots of mean calibration deviation by cell-line context (K562 vs. non-K562 datasets). Two-way Type-II ANOVA main effect f - $p = 0.0064$, $\eta^2 = 0.120$; $n = 80$ (predictor, dataset) cells (36 K562, 41 non-K562).

ate finding: K562-context cells have systematically smaller calibration deviation than non-K562-context cells, controlling for predictor capacity in the two-way Type-II ANOVA.

3.3. Pre-registration ablation

On the Phase 1 snapshot, the locked $p < 0.05$ threshold passes only one dataset for H_1 . A counterfactual relaxation $\{p < 0.10 \wedge n_{\text{pass}} \geq 2\}$ admits a second passer (Tahoe, $p = 0.066$) and would flip the disposition; the lock blocks the flip. On Phase 2D the relaxation is unnecessary — H_1 passes under the locked rule — but the Phase 1 flip stands as evidence of the threshold-shopping bias the lock prevents.

4. Discussion

Foundation models are the worst-calibrated family. A frozen encoder with a ridge head loses the per-gene variance structure needed to match full Perturb-seq distributions; advantage on supervised embedding-to-label tasks does not transfer to per-population sampling. The result is consistent with Ahlmann-Eltze et al. (2025); Csentesi et al. (2025); Kedzierska et al. (2025) under their accuracy metrics; we add a coverage criterion. This does not settle the full-fine-tune question: the transfer-learning mode we evaluate is the mode practitioners adopt when downstream labels are scarce.

Cell-line context is a real covariate. $\eta^2 = 0.120$ on the cell-line factor exceeds the pre-registered threshold. A plausible mechanism is K562-weighted training corpora biasing the learned variance structure toward the K562 marginal, which then propagates through frozen-encoder ridge heads. A targeted experiment would re-fit each predictor on subsamples of a single source corpus, holding to-

tal cell count fixed, and compare calibration deviation between K562 and non-K562 evaluation contexts.

Capacity dominates within F_3 but not across families.

Within the transformer family, calibration deviation increases monotonically with parameter count on the four H_1 passing datasets (Fig. 2). Across families, the family effect dwarfs the within-family capacity slope; parameter count is therefore a within-family predictor of calibration and architecture family is the dominant between-family predictor.

4.1. Limitations and future work

The benchmark reports marginal not conditional coverage; distribution-free conditional coverage is impossible (Vovk, 2012). Seven of the eight datasets are cancer or transformed cell lines, and the mouse ex-vivo slice is the only exception. F_3 predictors are evaluated in frozen-encoder mode; full perturbation fine-tune is left to future work. Hypotheses H_1 , H_{1b} , and H_{2b} are individually underpowered at their target effect sizes (Appendix C). Heavy-weight F_3 runs are restricted to six datasets per the pre-reg budget cap. Concrete next steps: re-evaluate F_3 under full `scgpt.Trainer.train_perturb` fine-tune on the same six datasets and report $\Delta\bar{d}$; re-run on Replogle Genome-Wide Perturb-seq to test H_3 replication at larger n ; add CellFM once a working PyTorch port is published; and replace the corpus-diversity index with a TF-IDF cell-type-coverage measure to re-test H_{2b} .

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A. Methods

Software environment. Python 3.11 (CPU) / 3.10 (CPA + scvi-tools subset). Key version pins: torch 2.4.1 (torch 2.3.1 in the scGPT image for torchtext 0.18.0 ABI compatibility), scvi-tools<1.4, cell-gears 0.1.2, scgpt 0.2.4, transformers>=4.40,<4.46. The full image graph (one per heavyweight predictor family) lives in `scripts/modal_launch.py`.

Hardware. Heavyweight runs use Modal A100 GPUs (one per (predictor, dataset), 24–32 GB host RAM). Lightweight

runs use a local M-series Apple Silicon CPU. Modal budget cap \$3000.

Preprocessing. `confpert.data._load_perturb_ds:` drop cells with < 200 UMI; drop genes detected in < 5 cells; `normalize_total(target_sum=1e4); log1p; highly_variable_genes(n_top=512)`. Top 50 perturbations by cell count are retained (29 for Datlinger). Random seed 42.

Hypothesis tests. H_1 : Spearman per dataset between $\log_{10}(\text{param count})$ and mean calibration deviation; pass if $\rho > 0.5$, $p < 0.05$ (permutation, $n_{\text{perm}} = 10\,000$), on at least 3 of 4 main datasets. H_{1b} : same with training-corpus size, $\rho < -0.5$. H_2 : two-way Type-II ANOVA on calibration deviation with factors (cell-line context, log param count); pass if main-effect f - $p < 0.0167$ (Bonferroni over 3 family tests) and $\eta^2 > 0.10$. H_{2b} : Spearman with log corpus diversity, $\rho < -0.5$, $p < 0.0167$, on at least 7 of 10 datasets. H_3 : Kruskal-Wallis across the three families, $p < 0.0167$ and Cliff's $\delta > 0.50$ between best and worst.

Bootstrap CIs and power. Wilson 95% binomial intervals per (predictor, dataset, score, α) cell; counts and intervals are in `phase2d_report_phase2.json:bootstrap_ci_table`. Power estimates use Monte Carlo ($n_{\text{sims}} = 2000$) for Spearman / Kruskal and `statsmodels.stats.power.FTestAnovaPower` for the ANOVA.

Reproduction.

```
git checkout <commit-SHA>
pip install -e .
python -m confpert.cli phase2d \
  --prereg ../preregistration_v2.yaml \
  --results baselines/results.json \
  --out ../phase2d_report_phase2.json
```

B. K_1 results (sidecar pointer)

Per-cell K_1 results (651 rows plus Wilson 95% intervals) live in the JSON sidecar at `paper_neurips_dnb_2026/phase2d_report_phase2.json` under the `bootstrap_ci_table` key. Each row is indexed by (predictor, dataset, score, α) and reports held-out perturbation count, observed coverage $\hat{p}$, and CI bounds.

C. Power analysis

Test	n	target effect	est. power
H_1 per-dataset Spearman	14	$\rho = 0.5$	0.41 (under)
H_{1b} cross-dataset Spearman	10	$\rho = -0.5$	0.25 (under)
H_2 two-way ANOVA	120	$\eta^2 = 0.10$	0.81
H_{2b} per-dataset Spearman	14	$\rho = -0.5$	0.26 (under)
H_3 Kruskal-Wallis (3 fam.)	120	$\delta = 0.5$	0.92

H_1 , H_{1b} , and H_{2b} are individually underpowered at their target effect sizes; the headline H_1 disposition is driven by larger observed effects on the passing datasets ($\rho \in [0.59, 0.71]$). H_2 and H_3 are independently well-powered.